

ear drum

Slender resistant elastic membrane; it vibrates when sound waves are received from the auditory canal, then transmits the waves to the ossicles.

auditory ossicles

The smallest bones in the human body, held in place by several muscles and ligaments; they amplify the vibrations of the ear drum.

posterior semicircular canal

Vertical canal parallel to the temporal bone; it monitors head movements to ensure that equilibrium is maintained.

superior semicircular canal

Vertical canal perpendicular to the temporal bone; it monitors head movements to ensure that equilibrium is maintained.

lateral semicircular canal

Horizontal canal; it monitors head movements to ensure that equilibrium is maintained.

vestibular nerve

Nerve transmitting messages related to equilibrium to the brain; it emanates from the vestibule and the semicircular canals.

cochlear nerve

Nerve transmitting auditory messages collected in the cochlea to the brain. The cochlear and vestibular nerves join to form the auditory nerve.

vestibule

Bony structure into which the three semicircular canals open; with these canals, it is responsible for equilibrium.

cochlea

Bony structure intended for hearing; it receives vibrations from the ossicles and transforms them into nervous impulses before transmitting them to the brain.

auditory ossicles

Each of the three small interarticulated bones of the middle ear that amplify the vibrations of the ear drum and transmit them to the internal ear.

Eustachian tube

Tube connecting the middle ear to the nasopharynx; it allows outside air to pass through, thus equalizing air pressure on both sides of the ear drum.

incus

Auricle of the middle ear articulating with the malleus and the stapes.

malleus

Auricle of the middle ear transmitting vibrations to the incus from the ear drum (to which it is attached).

stapes

Auricle of the middle ear transmitting vibrations from the incus to the internal ear; at about .15 in long, the stapes is the smallest bone in the body.

